

Eco Clean – Scheduling Solution

Progress Report 2 – Upul Atapattu

1. Summary

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1.1. Project Requirements Gathering

1.1.1. Changes to the Original Proposal

- Instead of Mobile App to be used by the staff, a PWA is expected to be implemented.
- As AI integration value addition, approximate travel time from staff location to client location to be calculated and optimized to support better appointment management.
- PWA to track Staff Time Tracking, with GPS-based clocked in/out, count down timer, job viewing and reminders.
- Principle web-application to accommodate and other staff related interfaces to update their availability, apply leave 2 weeks prior, submit their timesheets and view the paystubs. The staff app should be connected to Google Maps for navigation.
- Email notifications to be integrated with resend

1.2. Deployment in Vercel

In order to deploy the app with cloud-based functionalities, the following were done:

- DB migrated to Supabase
- Vercel Deployment

Please note that Vidarshan has taken lead on this front as initial attempts to resolve build failures in Vercel were not successful.

1.3. DB Schema Change

At Midterm, the database schema supported only the following interfaces and related functions.

- User management
- Appointment creation
- Job creation
- Assignment logic

Post-midterm, the database schema is updated to accommodate the staff related functionalities such as staff profile creation and updating, availability update, time-sheet submission and payroll functions.

The latest DB Schema demonstrated in Mermaid are as follows:[Click the image if it is not clear]

CLIENT			
String	id	PK	
String	title		NULLABLE
String	firstName		
String	lastName		
String	companyName		NULLABLE
String	email		
String	phone		
String	preferredContact		
String	leadSource		NULLABLE
DateTime	createdAt		
DateTime	updatedAt		

ADDRESS			
String	id	PK	
String	clientId	FK	
String	street1		
String	street2		NULLABLE
String	city		
String	province		
String	postalCode		
String	country		
Boolean	isPrimary		
Boolean	isBilling		
DateTime	createdAt		

JOB			
String	id	PK	
String	title		
String	jobType		
String	clientId	FK	
String	addressId	FK	
Boolean	isAnytime		
String	visitInstructions		NULLABLE
DateTime	createdAt		
DateTime	updatedAt		

APPOINTMENT			
String	id	PK	
String	jobId	FK	
DateTime	startTime		
DateTime	endTime		
DateTime	appointmentStatus	status	
Boolean	reminder1dSent		
Boolean	reminder5dSent		
DateTime	createdAt		
DateTime	updatedAt		

RECCURRENCE			
String	id	PK	
String	jobId	UNIQUE FK	
String	frequency		
Int	interval		
DateTime	endsOn		NULLABLE
String	endType		
Int	endsAfter		NULLABLE
DateTime	createdAt		

USER			
String	id	PK	
String	name		
String	email		UNIQUE
String	password		
Role	role		
DateTime	createdAt		

STAFF			
String	id	PK	
String	userId	UNIQUE FK	
Boolean	isActive		
Float	hourlyRate		NULLABLE
DateTime	createdAt		
DateTime	updatedAt		

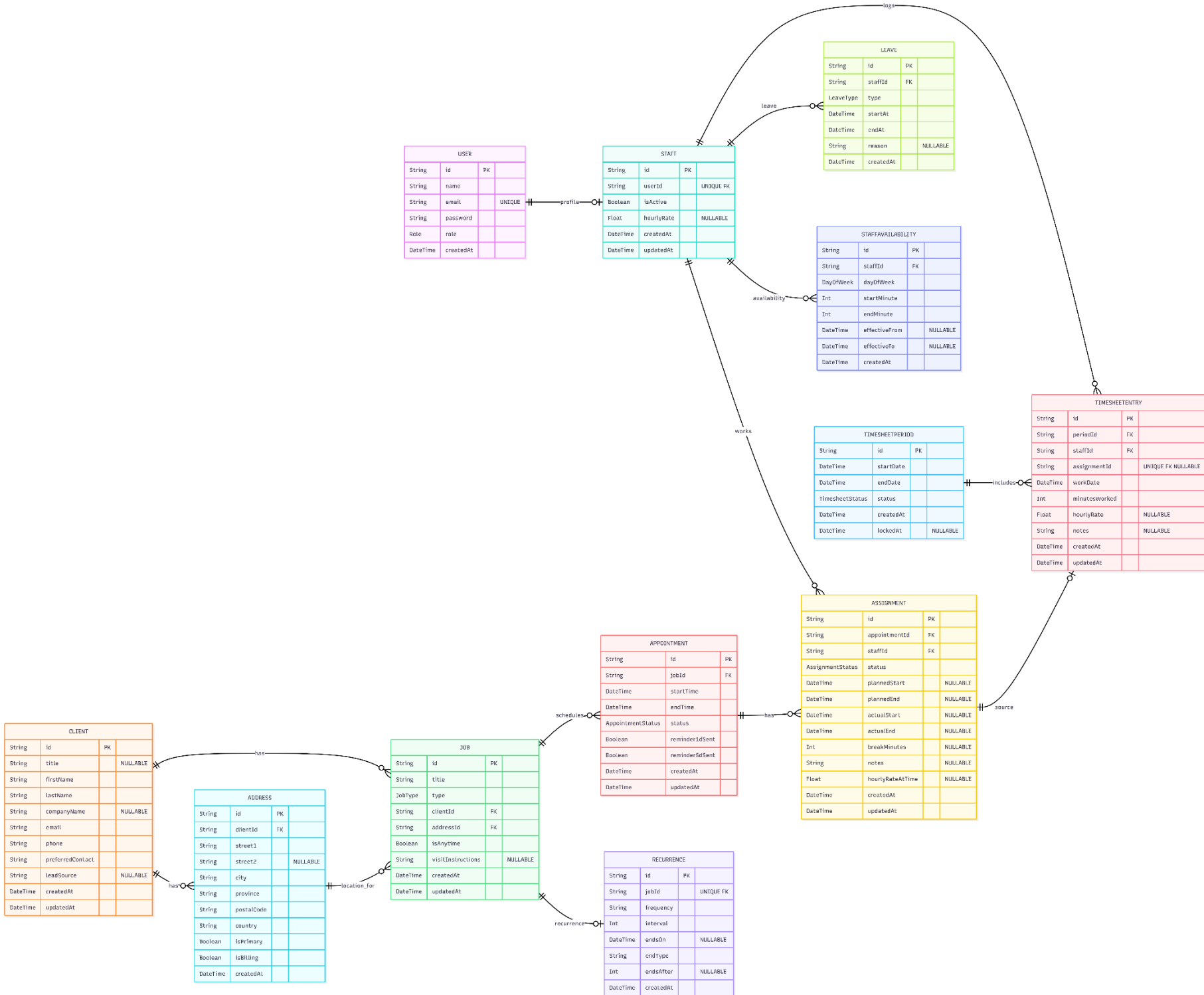
LEAVE			
String	id	PK	
String	staffId	FK	
LeaveType	type		
DateTime	startAt		
DateTime	endAt		
String	reason		NULLABLE
DateTime	createdAt		

STAFFAVAILABILITY			
String	id	PK	
String	staffId	FK	
DayOfWeek	dayOfWeek		
Int	startMinute		
Int	endMinute		
DateTime	effectiveFrom		NULLABLE
DateTime	effectiveTo		NULLABLE
DateTime	createdAt		

TIMESHEETPERIOD			
String	id	PK	
DateTime	startDate		
DateTime	endDate		
TimesheetStatus	status		
DateTime	createdAt		
DateTime	lockedAt		NULLABLE

ASSIGNMENT			
String	id	PK	
String	appointmentId	FK	
String	staffId	FK	
AssignmentStatus	status		
DateTime	plannedStart		NULLABLE
DateTime	plannedEnd		NULLABLE
DateTime	actualStart		NULLABLE
DateTime	actualEnd		NULLABLE
Int	breakMinutes		NULLABLE
String	notes		NULLABLE
Float	hourlyRateAtTime		NULLABLE
DateTime	createdAt		
DateTime	updatedAt		

TIMESHEETENTRY			
String	id	PK	
String	periodId	FK	
String	staffId	FK	
String	assignmentId	UNIQUE FK NULLABLE	
DateTime	workDate		
Int	minutesWorked		
Float	hourlyRate		NULLABLE
String	notes		NULLABLE
DateTime	createdAt		
DateTime	updatedAt		



1.4. DB Prisma Migration

After the new database schema, the newly added relationships and entities to the prisma schema design was successfully migrated. We have decided to keep two separate supabase schema, until we fully test the functionality of the new changes brought.

1.5. API Endpoint Creation for Staff Related Areas

API routes were created to support staff-related functionalities for the following areas:

- Staff profile creation and update
- Availability updates
- Leave request submission
- Timesheet submission

1.6. Commencement of Connecting the API with the UI Components

Since the test data was inserted to the newly created database, the process of connecting backend API endpoints with frontend UI components has commenced and currently in progress.

2. Worklogs

Date	Hours	Description
15th January 2026 2:30am-3:30am	1.5 hour	Prepare the project briefly after studying the company website and the project scope.
20th January 2026 1:30pm-2:00pm	0.5 hours	Had the initial meeting with Stephanie Riddel to understand the scope of the Riipen project.
24th January 2026 4:00pm - 5:00pm	1 hour	Researched on existing scheduling solutions (ADP) and what it currently serves, what are their capabilities to understand the current market.
25th January 2026 3:00pm – 4.30pm	1.5 hour	I studied and reviewed the section that Vidarshan has done on the technologies and the development tools.
25th January 2026 8:00pm – 8.30pm	0.5 hours	Prepared the project timelines and deliverables
26th January 2026 5:00pm – 8:30pm	3.5 hours	Worked on the project proposal to match the project briefly, giving background, project scope, benefits and justification and finalized.
29th January 2026 10:00pm – 11:00pm	1.5 hours	Cloned the Github repo and reviewed the code that Vidarshan had initiated. However, no contribution to the code was done.
30th January 2026 9:45pm – 11.30pm	1.75 hours	Discussed with Vidarshan on the wayforward in terms of the implementation. Discussed on few brainstorming ideas for initial design for the mobile app (Based on MiHCM) and web portal (based on ADP, timesheets used by Douglas Employees portal)
31st January 2026 3:00pm – 4:00pm 5:30pm – 6:30pm	2 hour	Started a Udemy course on React as my fullstack understanding is not enough to do a proper vite coding job. Covered the state management implementation and building form components with state.
1st February 2026 10:30am – 11:00am 12:00pm – 1:30pm 3:30pm – 4:30 pm 6:30pm – 7:00 pm	3.5 hours	Continued my learning on Udemy platform on component library development such as creating button, input and modal components.
2nd February 2026 9:00 pm – 10:00 pm 11:30pm – 12:30 am	2.0 hours	Covered styling system implementation includes setting up CSS modules and creating responsive layouts.

3rd February 2026 1:30pm – 2:30pm	1 hour	Had a google meet meeting with Stephanie Riddelle to talk about the business model, her requirement and the current process of scheduling.
4th February 2026 12:00am – 1:00am 2:00 am – 3:00am	2 hours	Continued my learning on Udemy platform on routing implementation topics such as setting up route structures, implementing protected routes for different user roles, and creating reusable navigation components.
5th February 2026 12:00 am – 1:30pm	1.5 hours	Reviewed the video call recording for better understanding of the scheduling process and the requirement. Based on Stephanie's recommendation, registered for www.getjobber.com as she preferred the user interface design and the workflow is better suited for her business.
7th February 2026 8:00 pm – 9:00pm	1.5 hours	Set up the project in my local machine with Vidarshan's help with the necessary env file for postgre integration.
8th February 2026 8:00 am – 9:00 am 2:00 pm – 3:00 pm 4:00 pm – 4:30 pm	2.5 hours	Continued my learning on Udemy platform on React useEffect topics including API integration patterns, loading and error state management, implementing theme selectors for light/dark mode functionality.
9th February 2026 7:00pm – 8:00pm	1 hour	Learnt about NextJS from YouTube, covering, why it is different from react, server-side rendering and API routes for backend integration for NextJS 15+.
11th February 2026 10:00pm – 11:00pm 12:00am – 1:00am	2 hours	Drafted improved UI/UX wireframes for admin dashboard based on ADP and Jobber systems.
13th February 2026 8:30 am – 9:30 pm 6:00pm – 7:30pm	2.5 hours	Ran into an issue with the development set-up due to postgre credentials mismatch. Had to remove the postgre and reinstall it in order to fix it. This has resulted in the previous data dump not integrating. Therefore, had tried various options 1. Tried creating another admin account using postman [which did not go as planned]. 2. Therefore, disabled the authentication logic to move forward with the development for a separate workbranch.

14th February 2026 8:30 am – 9:30 pm	1 hours	Since, my development branch got a warning, “The "middleware" file convention is deprecated. Please use "proxy" instead.”, worked on moving to the proxy file convention.
15th February 2026 8:30 am – 9:00 am 3:00 pm – 3:45 pm 5:00 pm – 5:30 pm	1.75 hour	I have started on the design of the staff related interfaces, specifically leave management and timesheets related interfaces.
16th February 2026	2 hours	I started implemented the frontend UI related elements of staff related interfaces, with the help of ChatGPT.
18th February 2026 4:00 pm – 5:00 pm	1 hour	Continued working on implementing the UI related elements that I designed and finished the two interfaces.
21st February 2026 9:00 pm – 10:00 pm	1 hour	Configured Supabase PostgreSQL hosting and environment variables get ready for Vercel deployment
22nd February 2026 8:30 am – 9:45 am 3:00 pm – 3:45 pm 5:00 pm – 6:30 pm 8:00pm – 9:00 pm	3.5 hour	After setting up the prerequisites for Vercel, attempted Vercel deployment; encountered various build-time strict TypeScript errors from my end. Eventhough significant time was spend for debugging this, could not get the deployment to build.
23rd February 2026 4:00 pm – 5:30 pm 6:30 pm -7.00 pm	2 hours	Started on improving the DB design, so, we can improve on facilitate the staff leave, tracking work logs which is needed to do timesheets and payroll calculations.
24th February 2026 4:30 pm – 5:30 pm 6:00 pm -8:00 pm 12:30 pm-1:00 pm 1:45 pm -2:15 pm	4 hours	Prepared the installation guidelines in the repo. Prepared the Mid Term Report. Recorded the Video Edited the Video and Submitted the Files. Since I have difficulty in troubleshoot with the Vercel deployment due to build errors, the Repo was transferred to Vidarshan for advance troubleshooting.
25th February 2026 5:00 pm – 6:30 pm 8:00 pm – 9:30 pm 11:30pm – 12:00 am	4 hours	Designed and created the “Enter Availability” interface and code was generated using ChatGPT. I have changed certain areas to match my design and the other designs. Started renewing the database schema to accommodate the Staff, Leave, Availability, and Timesheet entities. Extra

		careful not to change the existing tables/entities that are already linked with our deployment. In the exiting tables, only the relationship was changed. In some areas, duplications were kept as it is, being cautious about the implemented code.
26th February 2026 3:00 pm – 4:30 pm 8:00 pm – 8:30 pm 11:00 pm – 12:00am	3 hours	Updated Prisma schema to reflect improved staff-related entities and relationships. Started working on creating API endpoints looking at Vidarshan's code and getting help from Gemini. Completed API end points for staff and leave.
27th February 2026 10:00 pm – 10:30 pm 3:00 pm – 4:00 pm 11:00 pm – 11:30pm	2 hours	Midterm Evaluation and discussed about the way forward with Vidarshan looking at the remaining requirements. Kept a separate supabase database due to migrations. Continued on creating API end points for staff related tables. Specifically, for time-sheet entry and timesheet-period.
28th February 2026 4:00 pm – 5:00 pm 10:30 pm – 11:30 pm	2 hours	Started filling the database with test data with postman, where there were few unsuccessful entries due to foreign key violation. Later figured out the order of adding rows.
1st March 2026 8:30 am – 9:30 am 2:30 pm – 4:00 pm 5:30 pm – 6:30 pm 7:30pm – 8:30pm	4 hours	Completed the generation of testdata before moving to integration of interfaces to the APIs. Started integrating Staff Profile to its related data. (Even with the AI, the debugging took bit of time). Committed the changes I have done for the Upul_staff branch in GitHub.
2nd March 2026 5:00 pm – 6:15 pm 8:00 pm – 9:30 pm	2.75 hours	Integrated Staff Availability interfaces with its API point and to retrieve the dynamic rendering of availability records from database. Worked on creating the progress report and updating the AI section and worklogs.
Total	67.25 hours	

3. AI Usage

Please note that only the principal prompt is included in the table. In certain cases, I had put the entire error message or code, in case I was not able to pinpoint the error/issue.

AI	Prompt	Value Addition
ChatGP T, Free	I'm trying to create a scheduling web-app. I want create a 24*7 calender in HTML, so I can later use with react, how would you guide me to do it. Let me show it part by part	Created for learning purposes and to create JSX components if necessary to implement in react.
ChatGP T, Free	how to use https://fullcalendar.io/ for do a schedule calendar	For brainstorming purposes
ChatGP T, Free	when I'm trying to do scheduling web app with next js. What are the things that I should learn? I have a grasp of react. What is different in NextJs and what are the areas I should expand?	Learning purposes
ChatGP T, Free	how to setup my postgre env file	To troubleshoot in the development environment
Gemini, Free	how to deal with △ The "middleware" file convention is deprecated. Please use "proxy" instead. Learn more: https://nextjs.org/docs/messages/middleware-to-proxy × The file "./middleware.ts" must export a function, either as a default export or as a named "middleware" export. ?	To troubleshoot in the development environment
ChatGP T, Free	Would you help me to resolve the following error in the terminal? npm error code ERESOLVE npm error ERESOLVE could not resolve npm error While resolving: eco-clean-web@0.1.0 npm error Found: @mantine/core@8.3.14 npm error node_modules/@mantine/core npm error peer @mantine/core@"8.3.14" from @mantine/dates@8.3.14 npm error node_modules/@mantine/dates npm error @mantine/dates@"^8.3.14" from the root project npm error peer @mantine/core@"8.3.14" from @mantine/dropzone@8.3.14 npm error node_modules/@mantine/dropzone npm error @mantine/dropzone@"^8.3.14" from the root project npm error 1 more (the root project) npm error npm error Could not resolve dependency: npm error @mantine/core@"^8.3.13" from the root project npm error npm error Conflicting peer dependency: @mantine/hooks@8.3.15 npm error node_modules/@mantine/hooks npm error peer @mantine/hooks@"8.3.15" from @mantine/core@8.3.15 npm error node_modules/@mantine/core npm error @mantine/core@"^8.3.13" from the root project npm error npm error Fix the upstream dependency conflict, or retry npm error this command with --force or --legacy-peer-deps npm error to accept an incorrect (and potentially broken) dependency resolution. npm error	Fixed it by installing the Mantine version 8.3.14 based on the options provided by ChatGPT

ChatGP T, Free	What are the easier way to visualise the DB design from Prisma scheme?	To create DB design diagram
ChatGP T, Free	How can I generate a similar interface in NextJS (as shown in the image)? I'm hoping to use mantine	The same prompt was used to do vibecode and get the staff related interfaces. I had to manually the stylings, to match the profile I wanted.
ChatGP T, Free	What are the benefits of a PWA and limitations of it?	Learning purposes
ChatGP T, Free	Can a PWA send notifications?	Learning purposes
ChatGP T, Free	How can I create an installation instructions for my GitHub Repo. About the usage of the repository files. My project is a NextJS based web-app with Prisma integration. Do we have to ask them to setup postgre in their own machine?	For setting up the installation guidelines
ChatGP T, Free	RESEND_API_KEY="your-resend-key" UPLOADTHING_SECRET="your-uploadthing-secret" I'm still having trouble understanding these two? Do I create this in the ENV?	Learning purposes and to create the guidance on creating the .env file in the installation guide
ChatGP T, Free	Can you tell me how to do formatting in an MD file?	For my leaning purposes and style the installation Guide. With this response, I was able to do additional lists and code blocks to better format my installation guide.md in the GitHub Repo
ChatGP T, Free	When I try to link as below, nothing happens. am I missing something? <code><BigActionButton onClick={ () => console.log('/enter-availability')} ></code> Availability <code></BigActionButton></code>	For debugging, I was missing the navigation logic. Instead of router push, I was writing in the console
ChatGP T, Free	I want to create this interface in my enter-availability. Would you be able to help me with the coding? [Photo attached of the interface I created in powerpoint for enter-availability]	I used this to generate the code for the interface I designed.
ChatGP T, Free	What are the alternatives that we have for resend? for sending client notifications? Do we need to have a service provider for push notifications as well?	For brainstorming purposes
Gemini, Free	My colleague said I can fill the database with test data with postman without interface. Can you guide me how to do it? This is my StaffProfile table in Prisma Studio * id text hourlyRate float8	Using the instructions, I have generated the test data using postman

	postalCode text * userId text What should I do?	
Gemini, Free	I don't think the prerequisite API end point is not there. the following is my folder structure, where should I adjust the backend code for this prerequisite?	Figuring out the where to set up the API endpoints in our project
Gemini, Free	What does the error means? Property 'StaffProfile' does not exist on type 'Omit<PrismaClient<PrismaClientOptions, never, DefaultArgs>, "\$connect" "\$disconnect" "\$on" "\$transaction" "\$extends">'.ts(2339) HOw to resolve this?	To debug when I tried to fill the test data via postman. The api route had to be adjusted
Gemini, Free	I'm getting a 405Method Not Allowed	To debug when I tried to fill the test data via postman. The api route had to be adjusted. The required POST method was there, resulting this error.
Gemini, Free	Now it is missing required fields. I think we have to give the user fields input as well in the postman, right? can't you generate the jason body to create a new user and staff profile related to it?	To generate postman data for a test row in staff table
Gemini, Free	Let's do the staff-availability route.ts <pre> model StaffAvailability { id String @id @default(uuid()) staffProfileId String dayOfWeek Int // 0-6 (Sunday-Saturday) startMinute Int // Minutes from midnight (e.g., 480 for 8:00 AM) endMinute Int // Minutes from midnight effectiveFrom DateTime? effectiveTo DateTime? createdAt DateTime @default(now()) staffProfile StaffProfile @relation(fields: [staffProfileId], references: [id], onDelete: Cascade) } </pre> I need POST, GET, UPDATE methods	To generate the POST, GET, UPDATE Methods to create API end-points for staff-availability table
Gemini, Free	Do I have to create the route.ts logic inside of staff/[id] in order to retrieve the specific data?	To understand the right placing for update the API endpoint and create the required structure.
Gemini, Free	What is the following error? <pre> 143 status: 400 144 }); </pre>	For most of the test data generation, I was getting this error. This was due to

	<pre> 145 } → 146 const leave = await __TURBOPACK__imported__module__\$5b\$project\$5d2f\$lib\$2f\$prisma\$2e\$ts__\$5b\$app\$2 d\$route\$5d__\$28\$ecmascript\$29\$__["prisma"].leave.create(Foreign key constraint violated on the constraint: `Leave_staffId_fkey` at <unknown> (app\api\staff\id\leave\route.ts:60:38) at async POST (app\api\staff\id\leave\route.ts:60:19) 58 } 59 > 60 const leave = await prisma.leave.create({ ^ 61 data: { 62 staffId, 63 type, // e.g., "SICK", "VACATION" { code: 'P2003', meta: [Object], </pre>	Foreign key constraint violated. This was useful to decide on next steps
Gemini, Free	<pre> C:\Douglas\AppliedResearch\W26_4495_S2_UpulA\Implementation\eco-clean- web\.next\dev\server\chunks\[root-of-the-server]__296992d5_.js:177:167 174 } 175 }); 176 // 2. Create the assignment with a snapshot of the current rate → 177 const assignment = await __TURBOPACK__imported__module__\$5b\$project\$5d2f\$lib\$2f\$prisma\$2e\$ts__\$5b\$app\$2 d\$route\$5d__\$28\$ecmascript\$29\$__["prisma"].assignment.create(Foreign key constraint violated on the constraint: `Assignment_staffId_fkey` at <unknown> (app\api\assignment\route.ts:72:48) at async POST (app\api\assignment\route.ts:72:24) 70 71 // 2. Create the assignment with a snapshot of the current rate > 72 const assignment = await prisma.assignment.create({ ^ 73 data: { 74 appointmentId, 75 staffId, { code: 'P2003', meta: [Object], clientVersion: '7.4.1' } </pre>	Understanding the steps to handle the test data generation

	I get the following error when I put <pre>{ "appointmentId": "appointmentid1", "staffId": "43642b1b-9daa-44c8-990f-15b3cf7e26b6", "plannedStart": "2026-03-01T08:30:00Z", "plannedEnd": "2026-03-01T11:00:00Z", "notes": "Primary cleaner for this unit." }</pre> I have already created the appointmentid in the prisma studio	
Gemini, Free	Looking at my Prisma Schema, what is the order I should fill my test data in postman. <pre>generator client { provider = "prisma-client-js" } datasource db { provider = "postgresql" } model User { id String @id @default(uuid()) name String email String @unique role Role createdAt DateTime @default(now()) password String assignedAppointments Appointment[] @relation("AppointmentStaff") staffProfile StaffProfile? assignments Assignment[] leaves Leave[] timesheetEntries TimesheetEntry[] } model Client { id String @id @default(uuid()) title String?</pre>	1. User [Dependency] None 2. Client [Dependency] None 3. Job [Dependency] Client, Address 4. Appointment [Dependency] Job 5. Assignment [Dependency] Appointment, User 6. TimesheetEntry [Dependency] TimesheetPeriod, User, Assignment Gemini has provided this order, which made things easier to fill test data without violating the foreign key violation

```

firstName    String
lastName     String
companyName  String?
email        String
phone        String
preferredContact String
leadSource   String?
createdAt    DateTime @default(now())
updatedAt    DateTime @updatedAt
addresses    Address[]
notes        ClientNote[]
jobs         Job[]
}

model ClientNote {
  id      String @id @default(uuid())
  clientId String
  content String
  createdAt DateTime @default(now())
  client  Client @relation(fields: [clientId], references: [id], onDelete: Cascade)
}

model Address {
  id      String @id @default(uuid())
  clientId String
  street1 String
  street2 String?
  city    String
  province String
  postalCode String
  country String
  isPrimary Boolean @default(false)
  isBilling Boolean @default(false)
  createdAt DateTime @default(now())
  client  Client @relation(fields: [clientId], references: [id], onDelete: Cascade)
  jobs    Job[]
}

model Job {

```

```

id          String      @id @default(uuid())
title       String
type        JobType
clientId    String
addressId   String
createdAt   DateTime     @default(now())
updatedAt   DateTime     @updatedAt
isAnytime   Boolean      @default(false)
visitInstructions String?
appointments Appointment[]
address      Address      @relation(fields: [addressId], references: [id])
client       Client       @relation(fields: [clientId], references: [id])
lineItems    JobLineItem[]
notes        JobNote[]
recurrence    Recurrence?
}

model Appointment {
  id          String      @id @default(uuid())
  startTime    DateTime     @db.Timestamp(6)
  endTime      DateTime     @db.Timestamp(6)
  status        AppointmentStatus
  createdAt     DateTime     @default(now()) @db.Timestamp(6)
  completionSent Boolean      @default(false)
  reminder1dSent Boolean      @default(false)
  reminder5dSent Boolean      @default(false)
  jobId         String
  timeSpent      Int?
  completedAt    DateTime?
  job            Job         @relation(fields: [jobId], references: [id], onDelete: Cascade)
  images          AppointmentImage[]
  notes          VisitNote[]
  staff           User[]      @relation("AppointmentStaff")

  assignments Assignment[]

  @@index([startTime])
  @@index([endTime])
}

```

```

model Recurrence {
  id      String  @id @default(uuid())
  jobId   String  @unique
  frequency String
  interval Int
  endsOn   DateTime?
  endType  String
  endsAfter Int?
  job      Job    @relation(fields: [jobId], references: [id], onDelete: Cascade)
}

model JobLineItem {
  id      String  @id @default(uuid())
  jobId   String
  name     String
  quantity Int
  unitCost Float?
  unitPrice Float?
  total    Float?
  description String?
  job      Job    @relation(fields: [jobId], references: [id], onDelete: Cascade)
}

model JobNote {
  id      String  @id @default(uuid())
  jobId   String
  content  String?
  createdAt DateTime @default(now())
  job      Job    @relation(fields: [jobId], references: [id], onDelete: Cascade)
  images   JobNoteImage[]
}

model JobNoteImage {
  id      String  @id @default(uuid())
  noteId  String
  url     String
  note    JobNote @relation(fields: [noteId], references: [id], onDelete: Cascade)
}

```



```

model VisitNote {
  id          String    @id @default(uuid())
  content      String
  isClientVisible Boolean @default(false)
  appointmentId String
  createdAt    DateTime @default(now())
  appointment  Appointment @relation(fields: [appointmentId], references: [id], onDelete:
Cascade)
}

```

```

model AppointmentImage {
  id          String    @id @default(uuid())
  appointmentId String
  url         String
  fileKey     String?
  appointment Appointment @relation(fields: [appointmentId], references: [id], onDelete:
Cascade)
}

```

```

model StaffProfile {
  id          String    @id @default(uuid())
  userId      String    @unique
  user        User      @relation(fields: [userId], references: [id])
  postalCode  String?
  hourlyRate  Float      @default(0)
  availabilities StaffAvailability[]
}

```

```

model Assignment {
  id          String    @id @default(uuid())
  appointmentId String
  staffId     String
  status      AssignmentStatus @default(PENDING)

```

```

// Timing & Travel
plannedStart DateTime? // Includes estimated travel
plannedEnd   DateTime?
actualStart  DateTime? // Captured via staff clock-in

```

```

actualEnd DateTime? // Captured via staff clock-out
breakMinutes Int @default(0)

// Payroll Snapshots
hourlyRateAtTime Float? // Lock the rate in case staff gets a raise later

notes String?
createdAt DateTime @default(now())
updatedAt DateTime @updatedAt

// Relations
appointment Appointment @relation(fields: [appointmentId], references: [id], onDelete:
Cascade)
staff User @relation(fields: [staffId], references: [id])
timesheetEntry TimesheetEntry?
}

model StaffAvailability {
  id String @id @default(uuid())
  staffProfileId String
  dayOfWeek Int // 0-6 (Sunday-Saturday)
  startMinute Int // Minutes from midnight (e.g., 480 for 8:00 AM)
  endMinute Int // Minutes from midnight
  effectiveFrom DateTime?
  effectiveTo DateTime?
  createdAt DateTime @default(now())

  staffProfile StaffProfile @relation(fields: [staffProfileId], references: [id], onDelete: Cascade)
}

model Leave {
  id String @id @default(uuid())
  staffId String
  type LeaveType
  startAt DateTime
  endAt DateTime
  reason String?
  createdAt DateTime @default(now())

```

```

    staff User @relation(fields: [staffId], references: [id])
  }

model TimesheetPeriod {
  id      String      @id @default(uuid())
  startDate DateTime
  endDate  DateTime
  status   TimesheetStatus @default(OPEN)
  createdAt DateTime    @default(now())
  lockedAt DateTime?
  entries  TimesheetEntry[]
}

model TimesheetEntry {
  id      String @id @default(uuid())
  periodId String
  staffId  String
  assignmentId String? @unique
  workDate  DateTime
  minutesWorked Int
  hourlyRate Float?
  notes     String?
  createdAt DateTime @default(now())
  updatedAt DateTime @updatedAt

  period TimesheetPeriod @relation(fields: [periodId], references: [id])
  staff   User             @relation(fields: [staffId], references: [id])
  assignment Assignment?   @relation(fields: [assignmentId], references: [id])
}

enum Role {
  ADMIN
  STAFF
  CLIENT
}

enum JobType {
  ONE_OFF
  RECURRING
}

```

	<pre> } enum AppointmentStatus { SCHEDULED COMPLETED CANCELLED LATE } enum AssignmentStatus { PENDING EN_ROUTE ON_SITE COMPLETED SICK_LEAVE } enum LeaveType { SICK VACATION PERSONAL UNPAID } enum TimesheetStatus { OPEN SUBMITTED APPROVED LOCKED } Based on the current schema, can you tell me where should I start to do the test data creation? What is the order looking at the foreign constraints? </pre>	
Gemini, Free	How can I link my staff profile in the database for this interface <pre> export default function StaffProfilePage() { // Replace with data later (session / API) const staff = { </pre>	I have used the help from Gemini to start linking my database queries to my previously designed interfaces.

	name: 'Upul Atapattu', staffId: 'STF-0001', phone: '+1 (604) 555-0199', address: '12667 110A Avenue, Surrey, BC', postalCode: 'V3V 0A1', emergencyContact: 'Ayesha — +1 (604) 555-0123', }; }	
Gemini, Free	I believe I have to create route.ts for my staff/[id], right? inorder to fetch specific staff data? [Folder structure was attached as a picture]	I initially couldn't retrieve my data properly with a GET request, as I have not created an [id]/route.ts inside my staff folder in the api. I had to fix that error
Gemini, Free	What is cron? Why does it matter to sending email?	This was to understand some of Vidarshan's code on sending email. This give me some understanding on how cron is working for automating routine things

4. References

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